

Which Tail, and Probability of What?

AP Statistics · Unit 3 · Topic 3.6 · TEACHER KEY

CED TETHER

- SKILLS: Using Probability and Simulation
- **Skill 3.E** — Calculate a test statistic and find a p-value, provided conditions for inference are met.
- SKILLS: Statistical Argumentation
- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- ENDURING UNDERSTANDING: VAR-6 The normal distribution may be used to model variation.
- VAR-6.G Calculate an appropriate test statistic and p-value for a population proportion. [Skill 3.E]
- VAR-6.G.1 The distribution of the test statistic assuming the null hypothesis is true (null distribution) can be either a randomization distribution or when a probability model is assumed to be true, a theoretical distribution (z). VAR-6.G.2 When using a z-test, the standardized test statistic can be written: test statistic = (sample statistic - null value of the parameter) / (standard deviation of the statistic). This is called a z-statistic for proportions. VAR-6.G.3 The test statistic for a population proportion is: $z = (\hat{p} - p_0) / \sqrt{p_0(1 - p_0) / n}$. VAR-6.G.4 A p-value is the probability of obtaining a test statistic as extreme or more extreme than the observed test statistic when the null hypothesis and probability model are assumed to be true. The significance level may be given or determined by the researcher.
- ENDURING UNDERSTANDING: DAT-3 Significance testing allows us to make decisions about hypotheses within a particular context.
- DAT-3.A Interpret the p-value of a significance test for a population proportion. [Skill 4.B]
- DAT-3.A.1 The p-value is the proportion of values for the null distribution that are as extreme or more extreme than the observed value of the test statistic. This is: (a) the proportion at or above the observed value of the test statistic, if the alternative is $>$; (b) the proportion at or below the observed value of the test statistic, if the alternative is $<$; (c) the proportion less than or equal to the negative of the absolute value of the test statistic plus the proportion greater than or equal to the absolute value of the test statistic, if the alternative is $\neq$. DAT-3.A.2 An interpretation of the p-value of a significance test for a one-sample proportion should recognize that the p-value is computed by assuming that the probability model and null hypothesis are true, i.e., by assuming that the true population proportion is equal to the particular value stated in the null hypothesis.

Essential question: How do the alternative hypothesis and the null assumption determine a p-value and its meaning?

DOK-3 skill: 4.B · **FRQ pattern:** one-sided-vs-two-sided-p-value-interpret-the-condition

DOK rationale: Part (c) coordinates the research direction, null distribution, tail area, and conditional meaning of a p-value to adjudicate two interpretations and explain the inferential consequence of choosing the wrong tail.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- Which sample results are stronger evidence for a less-than alternative?
- What different research question would make 0.088 the relevant p-value?
- What must be assumed before the probability 0.044 is calculated?
- Is the random object in a p-value the fixed parameter or a future sample statistic?

[WARN] LOOK FOR

- Doubling a one-sided p-value merely because the sample proportion differs from the null value.
- Interpreting the p-value as the probability that H_0 or H_a is true.
- Describing exactly 98 supporters rather than 98 or fewer as the left-tail event.

SCORING PART (c) — E / P / I

Expected elements

- **judgment-1** (required) — Chooses Priya's left-tail probability near 0.044 and ties it to the fallen research question.
- **judgment-2** (required) — Interprets the probability as sample proportions at or below 0.49 in random samples of 200, assuming $p=0.55$.
- **judgment-3** (required) — Explains that Noah's doubled probability answers a two-sided question and does not measure the chance a hypothesis is true.

E	Meets all three checks with correct contextual reasoning; equivalent wording is acceptable.
P	Shows at least one substantive correct check, but has an omission or error that prevents E.
I	Shows none of the three checks with substantive correct reasoning; a choice alone is insufficient.

[STAR] THE PROBLEM — annotated

Suppose a city’s earlier planning model says 55% of households support expanded evening bus service. A researcher suspects support has *fallen*. From the city’s 20,000 households, a simple random sample of 200 finds 98 that support expansion, so $\hat{p} = 0.49$. Conditions for a one-sample z -test are met.

Priya: “For $H_0 : p = 0.55$ versus $H_a : p < 0.55$, use the left-tail p-value, about 0.044. If support is still 55%, about 4.4% of random samples of 200 would produce a sample proportion of 0.49 or lower.”

Noah: “Any difference from 0.55 counts, so double the tail to about 0.088. That means there is an 8.8% chance that the actual support proportion differs from 0.55.”

Sample counts

Group	Support	Other	Total
Households	98	102	200

FIRST TAKE — one sentence before you work the parts

Does fallen mean only a drop in support, or a change in either direction? Give one reason from the study.

Key: Not graded. The doubled value can sound more cautious; the finish should tie the tail to the question and the probability to repeated samples.

[BOOK] A p-VALUE CONDITIONS ON THE NULL (keep on the page)

For a one-sample proportion test, $z = \frac{\hat{p} - p_0}{\sqrt{p_0(1 - p_0)/n}}$. The p-value is the probability of a sample statistic **as extreme or more extreme** than the observed one, in the direction of H_a , assuming H_0 and the probability model are true. Use the left tail for $H_a : p < p_0$, the right tail for $H_a : p > p_0$, and both tails for $H_a : p \neq p_0$. A p-value is a probability about possible sample results under H_0 ; it is not the probability that a hypothesis is true. Printed normal table: $P(Z \leq -1.7056) = 0.044041$. For a symmetric normal curve, the two-sided area is twice the smaller tail area.

DOK 1 (a) State the null and alternative hypotheses and identify the tail direction required by the research question.

Key: Let p be the current proportion of all city households that support expanded evening bus service. The hypotheses are $H_0 : p = 0.55$ and $H_a : p < 0.55$. Because fallen means less than, the p-value is a left-tail area.

DOK 2 (b) Compute the one-sample proportion z -statistic and its one-sided p-value. Also compute the two-sided p-value Noah would obtain.

Key: The null standard deviation is $\sqrt{0.55(0.45)/200} \approx 0.035178$. Thus $z = (0.49 - 0.55)/0.035178 \approx -1.7056$. The one-sided p-value is $P(Z \leq -1.7056) \approx 0.04404$. Noah’s two-sided area is $2(0.04404) \approx 0.08808$.

DOK 3 (c) Choose the warranted p-value and interpretation. Use H_a and $\hat{p} = 0.49$ to define the tail event and null assumption, then explain what Noah’s doubled area answers and what his statement gets wrong.

Key: Priya is warranted. Because $H_a : p < 0.55$ and $\hat{p} = 0.49$, more extreme means $\hat{p} \leq 0.49$. Assuming $H_0 : p = 0.55$ and the model are true, that random-sample probability is about 0.044. Noah’s 0.088 answers the two-sided question $p \neq 0.55$. A p-value concerns possible samples under an assumed null, not

the probability that a parameter or hypothesis is true; Noah would falsely imply an 8.8% chance that the true support differs.